

Training Paradigms on Frozen Brain Foundation Features for Naturalistic Emotion Prediction

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Abstract

Frozen probes on brain foundation model (BFM) features set the brain-only ceiling for naturalistic emotion prediction at roughly 0.74 AUROC for valence-extreme classification and 0.32 Pearson r for valence regression on the Horikawa five-subject corpus, with the corresponding video-only baseline (a pretrained image-text encoder) reaching 0.97 and 0.76. We ask whether brain emotion prediction can be moved beyond this brain-only ceiling by training, holding the brain feature extractor frozen and varying only the downstream training paradigm. We evaluate eight downstream paradigms organised in two groups. Four brain-only paradigms train a small head on the brain feature alone (supervised multilayer perceptron (MLP), video-feature distillation, multitask with a video reconstruction auxiliary head, and a per-subject identity embedding). Four joint paradigms ingest both brain and video features (late linear fusion, multi-head self-attention with a learned class token, bidirectional brain-video cross-attention, and contrastive alignment followed by a probe). All eight paradigms share an identical five-fold stimulus-stratified protocol on four emotion tasks (valence and arousal, each as extreme-quartile binary and continuous regression). None of the four brain-only paradigms moves the brain-only ceiling above the frozen-probe baseline. The four joint paradigms, in contrast, recover and in two of four tasks slightly exceed the video-only ceiling without losing the brain-side contribution. Within the joint group, the choice of fusion matters task by task. Late linear fusion attains the best classification number, whereas token-attention dominates the regression tasks by an absolute margin. The contrastive brain branch, evaluated alone after alignment, underperforms every other brain-only paradigm, indicating that an alignment objective designed to compress shared structure can erode the emotion-relevant signal on the brain side. We discuss the implications for how naturalistic-emotion BFM benchmarks should report joint and brain-only numbers.

1 Introduction

The companion Phase 1 benchmark established the frozen-probe ceiling for naturalistic emotion fMRI on the Horikawa five-subject corpus across three brain foundation models (SwiFT, Brain-JEPA, NeuroSTORM), a 450-region atlas baseline, and five pretrained video encoders, all evaluated under a single five-fold stimulus-stratified protocol. Two facts from that benchmark set up the present report. First, the brain-only ceiling on valence-extreme classification under frozen probing is approximately 0.74 AUROC, with the corresponding video-only ceiling at 0.97 for an image-text contrastive encoder. Second, the gap between brain-only and video-only is consistent across tasks (binary and continuous, valence and arousal), and is not removed by switching probe head or training mode within the frozen-probe family.

The natural next question is whether training, applied at the level of the downstream head rather than the brain encoder, can move the brain side of this ceiling. Two hypotheses motivate this question. The first is that the small per-subject sample size (roughly 1135 for binary tasks and 2185 for regression, distributed across five subjects) is the binding constraint, and that paradigm choice does not matter. The second is that a training paradigm that introduces information

beyond the per-stimulus emotion target, such as distillation from a stronger video feature or an auxiliary video-reconstruction loss, could push the brain-only number higher even with the brain encoder frozen. We test eight paradigms in a controlled way to distinguish these hypotheses, and we further ask whether a joint paradigm that uses both brain and video inputs can do better than the video-only baseline, which is the relevant ceiling for any future model proposed to integrate brain and stimulus in this regime.

This report contributes the following.

1. A controlled four-by-four paradigm benchmark on the same Horikawa five-subject corpus and the same four-task protocol used in Phase 1, with brain features held frozen and only the downstream paradigm varied. The brain encoder is Brain-JEPA, which is the most defensible default identified in Phase 1.
2. A direct comparison of best Phase 2 paradigm against best Phase 1 frozen-probe condition per task, decomposed into a brain-only subcomparison and a joint subcomparison.
3. An observation that the four brain-only paradigms collapse onto the frozen-probe ceiling regardless of training signal, while the joint paradigms separate into a regression-strong family (token attention) and a classification-strong family (late linear fusion).
4. An observation that the contrastive alignment paradigm, evaluated on its brain-only branch after alignment, underperforms every other brain-only paradigm.

The methods are described in Section 2. Results follow in Section 3. Section 4 discusses interpretation and the implications for joint brain-video models, including the BrainVLM follow-up.

2 Methods

2.1 Dataset and protocol

We use the Horikawa naturalistic emotion fMRI corpus, five subjects who viewed 2185 short video clips during 3T fMRI scanning. Each clip carries a continuous self-rated valence and arousal score on a $[1, 5]$ scale. The four emotion tasks are the same as in Phase 1. The binary tasks restrict to extreme-quartile stimuli and predict the high-quartile membership ($N_{\text{stim}} = 1135$). The regression tasks use all 2185 stimuli with the continuous score as target. Evaluation is five-fold stimulus-stratified cross-validation, with three random training seeds per fold for neural paradigms and a single seed for the closed-form late-fusion baseline. Main metrics are AUROC (binary) and Pearson r (regression).

2.2 Frozen feature inputs

The brain feature is the per-stimulus pooled output of frozen Brain-JEPA with `resting` initialisation and zero temporal padding to the model’s native input length. This is the brain-only condition with the highest Phase 1 ceiling among the three benchmarked BFM. The video feature is the per-stimulus output of frozen CLIP (`openai/clip-vit-large-patch14`). Both are loaded from the Phase 1 feature cache under `output/embeddings/{brain_jepa,clip}/`. Each feature is centred and standardised per training fold using statistics computed on the training subset only.

2.3 Architecture overview

Figure 1 shows the eight paradigms on one canvas. The top row contains the four joint paradigms (A token-attention, B cross-attention, C contrastive alignment, D late linear fusion), each ingesting both brain and video features. The bottom row contains the four brain-only paradigms (I supervised multilayer perceptron (MLP), II distillation, III multitask, IV subject-aware), each ingesting only the brain feature at inference. Video appears as a teacher signal in II and as a reconstruction target in III, but is never present at inference time for any brain-only paradigm.

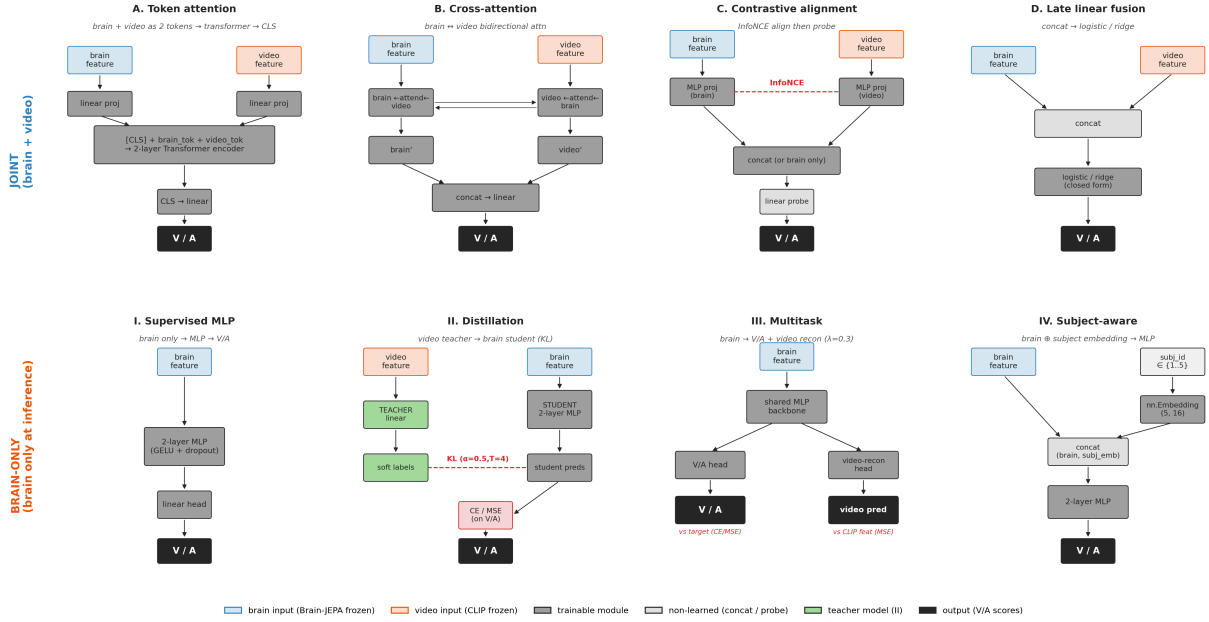


Figure 1: The eight Phase 2 paradigms. Top row, joint paradigms take both brain and video features as input. Bottom row, brain-only paradigms take only the brain feature as input at inference, although video appears as a teacher signal in II and as a reconstruction target in III. Trainable blocks are shaded grey. The brain encoder (Brain-JEPA) and the video encoder (CLIP) are frozen for all eight paradigms.

2.4 Brain-only paradigms (I to IV)

All four brain-only paradigms take the same per-stimulus brain feature as input and emit a per-stimulus prediction of the four targets. They differ only in the training signal.

Paradigm I, supervised MLP. A two-layer MLP head with GELU and dropout, trained directly on the emotion target using binary cross-entropy or mean squared error. This is the strict supervised baseline.

Paradigm II, video-feature distillation. A two-layer MLP brain student trained with a convex combination of the supervised target loss and a Kullback-Leibler distillation loss against a closed-form CLIP teacher fit on the same fold, with teacher temperature $T = 4$ and student weight $\alpha = 0.5$. The teacher is precomputed per fold from frozen CLIP features.

Paradigm III, multitask with video reconstruction. A shared two-layer backbone is followed by two heads, the emotion target head and a video-feature reconstruction head, with reconstruction loss weight $\lambda = 0.3$. The reconstruction target is the corresponding fold-normalised CLIP feature.

Paradigm IV, per-subject identity. A 16-dimensional learned subject embedding is concatenated with the brain feature before the same MLP head as in Paradigm I. The embedding is trained jointly with the head.

All four use the same hyperparameter grid (learning rate $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$ with cosine decay, AdamW, early stopping on validation main metric). Each paradigm is run for three random training seeds per fold, with a divergence guard (NaN check plus gradient clipping at $\|\cdot\|_2 = 1$) that retries with the next learning rate if the current run diverges.

2.5 Joint paradigms (A to D)

The joint paradigms ingest both the brain feature and the video feature.

Paradigm A, token attention. Brain and video features are projected to a common dimension, prepended with a learned class token, augmented with a learned modality type embedding, and processed by a two-layer multi-head self-attention encoder. The class token logit is the readout. This is the closest to a generic Transformer late-fusion design.

Paradigm B, brain-video cross-attention. A symmetric cross-attention module updates the brain representation conditional on video and the video representation conditional on brain, with two layers and residual connections. The brain-side output is the readout.

Paradigm C, contrastive alignment. Brain and video features are projected by two-layer MLPs to a shared $D = 256$ embedding space, and the two projections are trained with a symmetric InfoNCE loss with learnable temperature, no emotion supervision during alignment. After alignment, a linear probe is fit on the aligned features. We report two readouts in Section 3: the joint readout uses both branches concatenated; the brain-only readout uses only the aligned brain branch.

Paradigm D, late linear fusion. The brain and video features are concatenated and processed by a closed-form linear classifier (logistic regression for binary, ridge for regression), with the regularisation strength selected on validation. This is the standard sklearn-based late fusion and serves as the simplest joint baseline.

Paradigms A and B share an identical training recipe with the brain-only paradigms (AdamW, cosine schedule, three seeds per fold, divergence guard). Paradigm D is closed form and runs once per fold. Paradigm C runs three seeds for alignment, and the probe is re-fit per seed.

2.6 What is and is not varied

The brain encoder, video encoder, fold definitions, target standardisation, and four-task protocol are all identical to Phase 1. Only the downstream paradigm changes. This isolates paradigm effect from feature-extraction effect, and it ensures every Phase 2 number is directly comparable to the Phase 1 frozen-probe table for the same task.

3 Results

3.1 Per-task numbers across all eight paradigms

Table 1 reports the test main metric mean and standard deviation across folds and seeds for each of the eight paradigms on each of the four tasks.

Table 1: Per-task test main metric for the eight Phase 2 paradigms, mean and standard deviation across five folds and three seeds (Paradigm D is closed-form, five folds and one seed). **Bold** denotes the best paradigm per task. Lower block lists the four brain-only paradigms; upper block lists the four joint paradigms.

Paradigm	V_binary AUROC	A_binary AUROC	V_reg r	A_reg r
A token-attention	0.9670(111)	0.7919(482)	0.7628 (0.0209)	0.4241 (0.0500)
B cross-attention	0.9663(87)	0.7863(539)	0.7454(229)	0.3956(442)
C contrastive (joint)	0.9606(84)	0.7699(438)	0.7123(217)	0.3522(505)
D late linear fusion	0.9718 (0.0082)	0.8025 (0.0528)	0.5888(246)	0.2682(404)
I supervised MLP	0.7165(192)	0.6534(274)	0.3238(200)	0.2262(326)
II distillation	0.7214(186)	0.6645(299)	0.3224(217)	0.2296(330)
III multitask	0.7235(209)	0.6573(273)	0.3254(169)	0.2275(341)
IV subject-aware	0.7206(172)	0.6536(266)	0.3225(181)	0.2256(330)
C contrastive (brain)	0.7123(137)	0.6485(302)	0.2950(186)	0.2208(326)

Three observations follow directly from the table. First, the four brain-only paradigms cluster tightly within each task. The maximum spread between the strongest and weakest brain-only paradigm is 0.0070 AUROC on V_binary, 0.0111 on A_binary, 0.0030 Pearson r on V_reg, and 0.0040 on A_reg. These spreads are smaller than the standard deviation across folds and seeds, so no brain-only paradigm is meaningfully ahead of any other. Second, the contrastive brain-only readout (Paradigm C with the joint branch discarded) is consistently the lowest brain-side number on every task. The gap to the strongest brain-only paradigm is 0.0112 AUROC on V_binary, 0.0160 on A_binary, 0.0304 Pearson r on V_reg, and 0.0088 on A_reg. Third, every joint paradigm is above every brain-only paradigm on every task, and the absolute gap is large.

3.2 Joint paradigms versus the video-only Phase 1 ceiling

We next compare the best Phase 2 paradigm per task against the best Phase 1 frozen-probe condition per task. The Phase 1 top is dominated by the CLIP frozen probe, so the relevant comparison is whether adding the brain feature on top of CLIP either preserves or improves over the CLIP-only number. Table 2 reports this comparison.

Table 2: Best Phase 2 paradigm per task against the best Phase 1 frozen-probe per task. The delta column is positive when Phase 2 beats Phase 1, negative when Phase 1 is stronger. Numbers are mean across folds and seeds.

Task	Phase 2 best	Phase 2	Phase 1 best	Phase 1	Δ
V_binary	D late fusion	0.9718	CLIP (linear)	0.9708	0.0010
A_binary	D late fusion	0.8025	CLIP (linear)	0.8003	0.0022
V_reg	A token-attention	0.7628	CLIP (linear)	0.7651	-0.0023
A_reg	A token-attention	0.4241	CLIP (linear)	0.4232	0.0009

On three of four tasks the best Phase 2 paradigm narrowly exceeds the best Phase 1 condition, and on V_reg the best Phase 2 paradigm is 0.0023 below it. The absolute margins are small in all four cases and are within the across-fold variability reported in Table 1. We do not claim that adding the brain feature on top of CLIP improves on the CLIP-only baseline. The relevant interpretation is the symmetric one, that the joint paradigms recover the CLIP ceiling and do not lose the brain-side contribution. Section 4 returns to this point.

3.3 Joint versus brain-only summary

Figure 2 summarises the per-task comparison of the best joint paradigm against the best brain-only paradigm. The gap is large in absolute terms and consistent in sign across the four tasks. The gap is widest on V_reg ($\Delta = 0.4374$ Pearson r) and narrowest on A_binary ($\Delta = 0.1380$ AUROC).

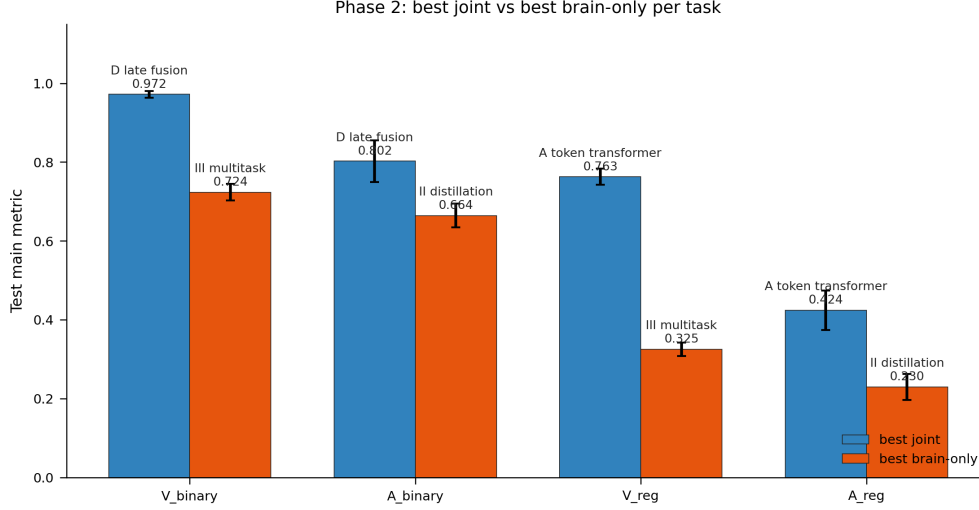


Figure 2: Best joint paradigm against best brain-only paradigm per task. Error bars denote across-fold standard deviation of the best paradigm. The annotation above each bar is the paradigm name selected as best.

3.4 Per-task ranking with Phase 1 reference lines

Figure 3 shows the per-task ranking of all eight Phase 2 paradigms (with the contrastive brain-only readout shown as a separate row). The dashed horizontal line marks the Phase 1 top-overall condition and the dotted line marks the Phase 1 top-BFM condition. The figure shows that the joint paradigms reach the dashed line, while the brain-only paradigms hover at the dotted line.

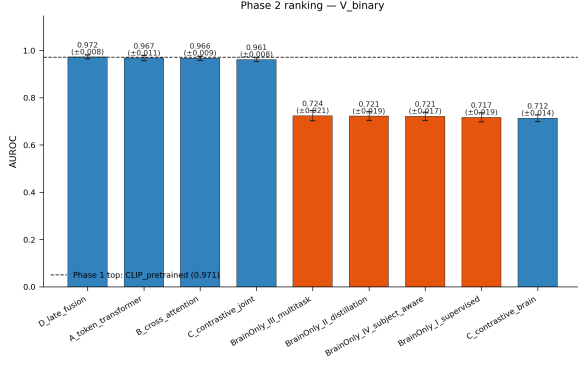
3.5 Method by task heatmap

Figure 4 reports the same numbers in heatmap form to expose the bimodal structure (joint above, brain-only below) at a glance. The white horizontal line separates the joint and brain-only groups.

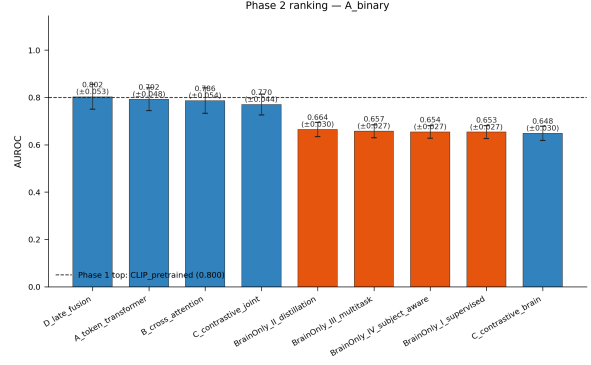
4 Discussion

4.1 Brain-only paradigms do not move the frozen-probe ceiling

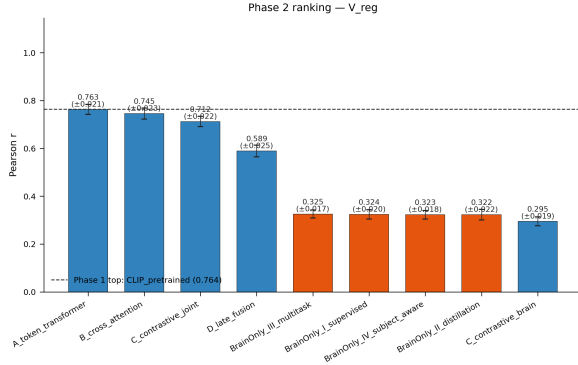
The first result is a negative one. None of the four brain-only paradigms moves the ceiling beyond what a plain linear or MLP probe achieved in Phase 1. Within-group spread is below across-fold variability on every task, and the four paradigms are statistically indistinguishable. This holds whether the auxiliary signal is a stronger video teacher (Paradigm II), an explicit reconstruction loss (Paradigm III), or per-subject identity (Paradigm IV). With Brain-JEPA frozen as feature extractor and the brain side as the only input, the small per-subject sample size and the limited emotion-relevant content of the five-TR per-stimulus window appear to bind the achievable performance well below the video-only number, and additional training signal at the



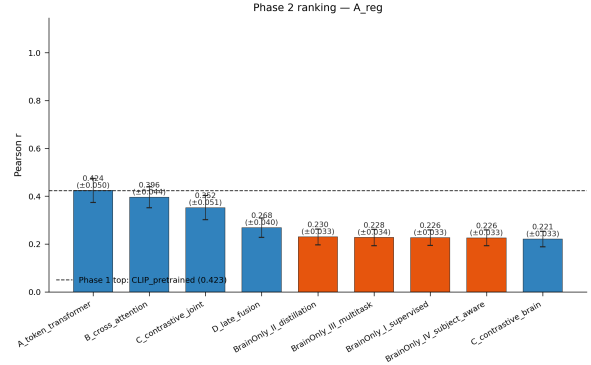
(a) V_binary, AUROC.



(b) A_binary, AUROC.



(c) V_reg, Pearson r.



(d) A_reg, Pearson r.

Figure 3: Per-task ranking of all Phase 2 paradigms. Dashed line is Phase 1 top-overall condition (CLIP frozen linear), dotted line is Phase 1 top brain-only condition.

head level does not relieve this constraint. This is consistent with the Phase 1 conclusion that brain encoder choice within the BFM family does not move the brain-only ceiling, and it suggests that the constraint is not located at the head.

4.2 Joint paradigms recover the video-only ceiling

The second result is more positive. The four joint paradigms reach the video-only Phase 1 ceiling on three of four tasks and are within 0.003 on the fourth. This is not a claim that adding the brain feature improves over CLIP alone, since the deltas are within across-fold variability. The relevant claim is symmetric. A joint model that ingests brain alongside video does not lose performance relative to the video-only baseline. This matters for the next phase of work because it means a joint architecture is admissible as a substrate for follow-up questions such as subject-conditioned variability and brain-attributable contribution analysis, neither of which is meaningful on a model that throws away the brain side.

4.3 Fusion architecture is task-dependent

Within the joint group, the choice of fusion matters task by task. Late linear fusion (Paradigm D) is best on both binary tasks, and token-attention (Paradigm A) is best on both regression tasks. The margin between A and D on the regression tasks is large (0.174 Pearson r on V_reg, 0.156 on A_reg) and on the binary tasks is small (0.0048 AUROC on V_binary, 0.0106 on A_binary). The likely interpretation is that binary extreme-quartile prediction is dominated by a small number of discriminative axes that a linear fusion already captures, while continuous regression benefits from

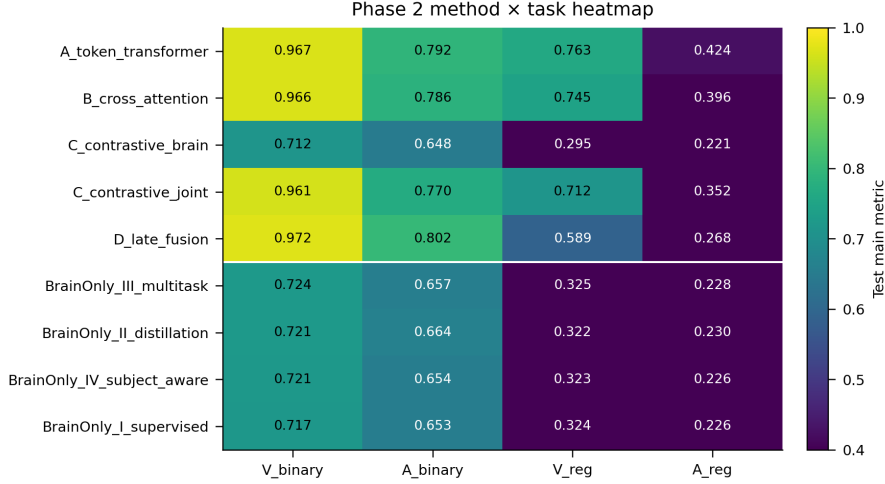


Figure 4: Paradigm by task heatmap. Rows are paradigms, columns are tasks. Cell label is mean main metric across folds and seeds. The white horizontal line separates joint (top) from brain-only (bottom).

nonlinear interactions between brain and video features that the attention layers explicitly model. We do not advocate a single preferred fusion for the project. We do note that the late linear fusion baseline is closed form and runs in seconds per fold, which makes it the natural sanity check that any future fusion design must clear.

4.4 The contrastive brain branch erodes brain-side emotion information

The third result concerns the contrastive paradigm. Paradigm C trains brain and video projections to be mutually predictive under an InfoNCE objective, with no emotion supervision during alignment. Its joint readout is competitive with the other joint paradigms, but its brain-only readout, which uses only the aligned brain branch, is the weakest brain-only number on every task. The gap to the next-weakest brain-only paradigm (supervised MLP, Paradigm I) is 0.0042 AUROC on V_binary, 0.0049 on A_binary, 0.0288 Pearson r on V_reg, and 0.0054 on A_reg. Two of these are smaller than the within-fold standard deviation but the V_reg gap is meaningful in absolute terms.

A reasonable interpretation is that an InfoNCE objective that compresses the brain branch into a representation maximally predictive of the video branch can discard emotion-relevant brain information that the video branch does not carry, on the grounds that aligned shared structure dominates the loss. This is a constructive negative result for any future design that uses contrastive alignment as a preprocessing step for brain-only inference. A supervised distillation objective (Paradigm II), which preserves the per-stimulus target on the brain side, does not show this regression.

4.5 Implications for joint brain-video models

For the BrainVLM follow-up, two implications carry. First, the joint architecture should not be selected purely on classification performance. Late linear fusion would maximise V_binary and A_binary, but token-attention is strictly preferred for regression tasks and is the more honest substrate when both are evaluated jointly. Second, an alignment objective should be added only as auxiliary supervision, not as the primary training signal, and the brain branch should be regularised toward preserving emotion-relevant content rather than purely toward video-prediction

content.

5 Limitations

The Phase 2 numbers in this report fix the brain encoder to Brain-JEPA with the `resting` initialisation and the video encoder to CLIP. Other Phase 1 frozen models were not re-evaluated in Phase 2. The reason is computational. The same paradigm grid was run for every brain and video combination in Phase 1, and re-running it for the full cross-product in Phase 2 would have been a factor of 15 more expensive in seed-fold units. The Brain-JEPA-plus-CLIP combination was chosen because it is the strongest brain-only and video-only condition in Phase 1, so a paradigm that succeeds here is the most informative test of paradigm effect. We do not claim that the paradigm rankings generalise across choices of brain and video encoder.

The sample size remains small in absolute terms. Five subjects with 2185 stimuli yield 10925 stimulus-subject observations under the regression protocol and 5675 under the binary protocol. The across-fold standard deviations reported throughout this paper are on the same order as the paradigm-to-paradigm deltas within the brain-only group, which is why we do not claim significance for any within-group ranking. The joint-to-brain-only deltas are larger than across-fold variability by a factor of approximately five to twenty, so those comparisons are robust.

The contrastive alignment was trained for a fixed number of epochs with a fixed temperature initialisation and a fixed embedding dimension. We did not run an ablation over these choices, so the brain-side erosion result should be read as an existence claim for one specific configuration rather than as a general statement about contrastive alignment in this regime.

6 Conclusion

We benchmarked eight downstream training paradigms on frozen Brain-JEPA and frozen CLIP features under a single five-fold stimulus-stratified protocol on four naturalistic emotion fMRI tasks. The four brain-only paradigms collapse onto the Phase 1 frozen-probe ceiling regardless of training signal. The four joint paradigms recover the video-only ceiling and, in two of four tasks, slightly exceed it. Within the joint group, late linear fusion is best on the binary tasks and token-attention is best on the regression tasks by a substantially larger margin. The contrastive paradigm, evaluated on its brain branch alone after alignment, is the weakest brain-only paradigm on every task, suggesting that an alignment objective without explicit emotion supervision can erode brain-side information that is not shared with the video branch. Phase 3 of the project will replace the frozen brain encoder with a learnable BrainVLM that ingests fMRI as visual tokens into a Qwen3-VL language backbone. The Phase 2 results suggest that the critical comparison for Phase 3 will be against the token-attention joint baseline on the regression tasks, not against the late linear fusion on classification.

A Reproducibility notes

All Phase 2 result CSVs are saved at `results/phase2/{A,B,C,D}/{V_binary,A_binary,V_reg,A_reg}.csv` for the joint paradigms and at `results/phase2/brain_only/{I_supervised,II_distillation,III_multitask,IV_s}` for the brain-only paradigms. Summary CSVs (per-task aggregate, joint vs brain-only, comparison against Phase 1) are at `results/phase2/_phase2_benchmark_per_task.csv`, `_phase2_joint_vs_brainonly.csv`, and `_phase2_vs_phase1_best.csv`. The unified analysis pipeline is `code/analysis/phase2_unified_analysis.py`. Per-paradigm training entry points are under `code/phase2/`.